

Smart Hospital Agentic AI System

Compliance & Emergency Monitoring Platform

Final Year Project — Detailed Technical Proposal
Multi-Agent AI Architecture for Safe Hospital Operations

LSTM/GRU Vitals

VIT Medical Imaging

SOP Compliance Engine

Real-Time Alerts



Vital
Agent

Disease
Agent

Vision
Agent

Resource
Agent

Compliance

1. Executive Summary

The **Smart Hospital Agentic AI** system is a next-generation healthcare intelligence platform built on a **multi-agent AI architecture**. Instead of relying on a single monolithic model, the system distributes specialised responsibilities across five purpose-built AI agents, each optimised for its domain. A central **Supervisor Agent** orchestrates collaboration and produces consolidated, safe recommendations.

The core research contribution is the **Compliance & Governance Layer**: no AI-recommended action — such as transferring a patient to the ICU, escalating an emergency, or adjusting medication — can be executed without first passing through an SOP (Standard Operating Procedure) verification engine. This transforms a capable AI system into a demonstrably *Safe AI* platform, an area of significant academic interest and real-world necessity.

Aspect	Detail
Domain	Smart Healthcare / Hospital Management
Architecture	Multi-Agent AI with Supervisor Orchestration
Core AI Models	LSTM, GRU, 1D-CNN, ViT, MobileViT, BERT
Key Innovation	AI Compliance & SOP Verification Layer
User Interface	Real-time web dashboard (MediGuard AI)
Academic Value	Publishable — extends multi-agent AI to healthcare
Datasets	MIMIC-III, PhysioNet, CheXpert (open-access)

2. System Architecture

The system follows a **hierarchical multi-agent pattern**. Patient data from IoT sensors, electronic health records (EHR), and imaging devices feeds into five specialist agents. Each agent runs its own AI model, produces a structured output, and passes it to the Supervisor Agent, which synthesises a final recommendation before the Compliance Agent validates it.

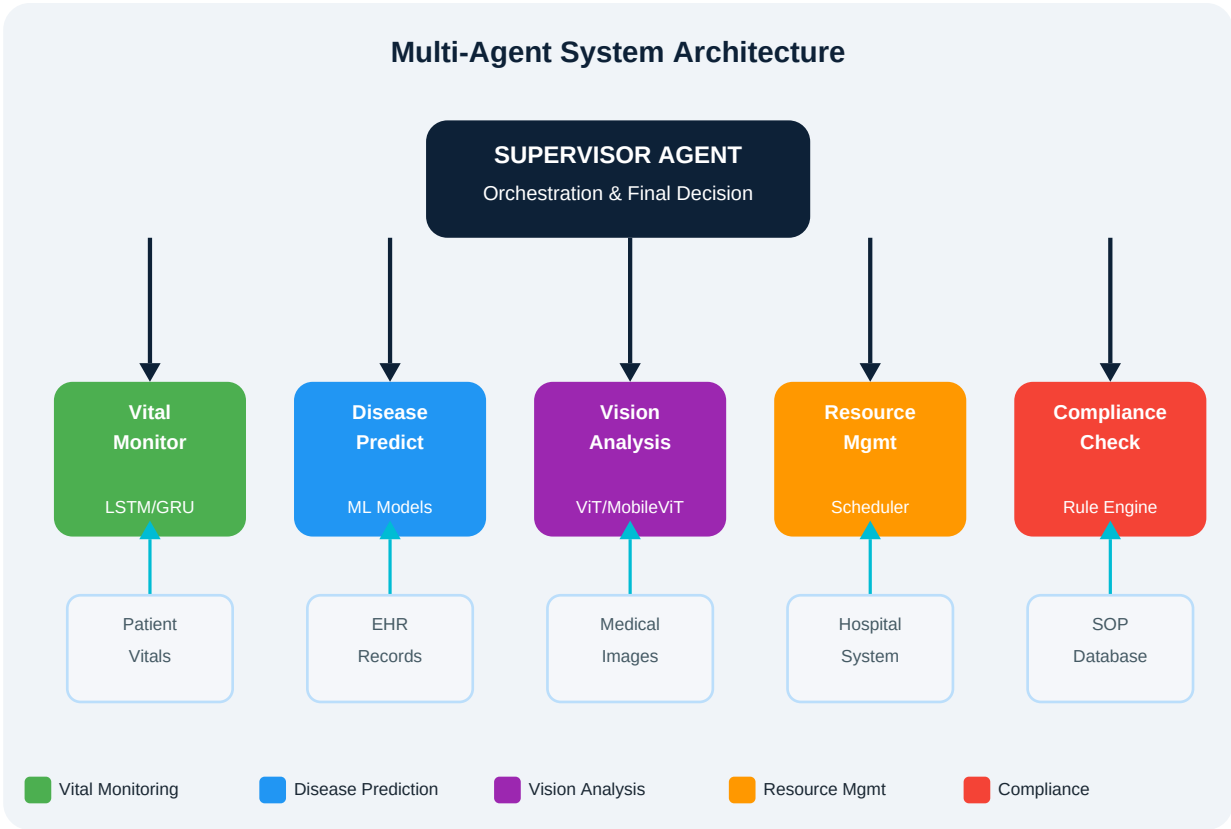


Figure 1 — Multi-Agent System Architecture

3. The Five Specialised Agents

Agent A — Vital Monitoring Agent	Model: LSTM / GRU (time-series)
Continuously reads patient vitals — heart rate, blood pressure, SpO2, temperature, and respiratory rate — from bedside IoT monitors. Uses a stacked LSTM/GRU recurrent network trained on ICU time-series datasets (PhysioNet) to detect anomalous trends, predict deterioration windows (up to 2 hours ahead), and raise pre-emptive alerts before a critical event occurs.	
• Inputs: IoT sensor streams, ECG, pulse oximeter, BP monitor • Model: Stacked LSTM + GRU with attention mechanism • Output: Anomaly flag, rolling trend report, ICU escalation probability (0–100%)	
Agent B — Disease Prediction Agent	Model: XGBoost Ensemble + Clinical Transformer
Aggregates lab results, medication history, patient demographics, and vital trends to predict the likelihood of sepsis, cardiac events, respiratory failure, and other acute conditions. Re-evaluates every 4 hours (or immediately after a vital alert). Generates a Risk Score (0–100) and an estimated time-to-critical-event window that clinicians can act on.	
• Inputs: EHR records, lab results, patient history, vital trend data • Model: XGBoost ensemble + lightweight clinical transformer • Output: Risk score, disease probability per category, recommended urgency level	

Agent C — Medical Vision Agent

Model: ViT / MobileViT (Vision Transformer)

Analyses medical images including X-rays, CT scans, MRI slices, and dermatology photographs. Uses a fine-tuned Vision Transformer (ViT-Base) to detect pneumonia, fractures, tumours, skin lesions, and other pathologies. A lightweight MobileViT variant enables deployment on hospital edge hardware. Outputs annotated heatmaps so radiologists can verify AI findings without anchoring bias.

- Inputs: DICOM images — X-ray, CT, MRI, skin photos
- Model: ViT-Base fine-tuned on CheXpert + ISIC datasets
- Output: Detected pathology label, confidence score, Grad-CAM heatmap overlay

Agent D — Resource & Appointment Agent

Model: Constraint Optimisation Scheduler

Handles bed allocation, ICU vacancy tracking, doctor shift scheduling, surgical theatre booking, and medical equipment assignment. Integrates with the hospital management system to prioritise resources by clinical urgency — a patient with a high risk score automatically jumps the queue for ICU bed assignment.

- Inputs: Hospital system API, live bed status, staff calendars
- Model: Constraint optimisation (OR-Tools) + priority queue
- Output: Resource allocation plan, bed assignment, appointment confirmation

Agent E — SOP & Compliance Agent

Model: Rule Engine + BERT Policy Parser

This is the **core research contribution** and the defining feature of this system. Before any agent recommendation is executed, the Compliance Agent checks it against a database of hospital SOPs, Ministry of Health regulations, drug interaction rules, and clinical governance guidelines. A BERT-based NLP model parses unstructured policy documents and converts them into machine-checkable rules. If any step is violated, the action is **blocked**, the responsible clinician is alerted, and a full explanation is logged in the audit trail.

- Inputs: Agent recommendations, SOP database, EHR audit trail, patient state
- Model: Rule engine + fine-tuned BERT for policy parsing
- Output: PASS / FAIL verdict, missing-step list, compliance audit log entry

4. Patient Journey & System Workflow

From the moment a patient arrives, the system continuously monitors, analyses, and coordinates across all agents. Every significant event triggers a handoff in the pipeline.

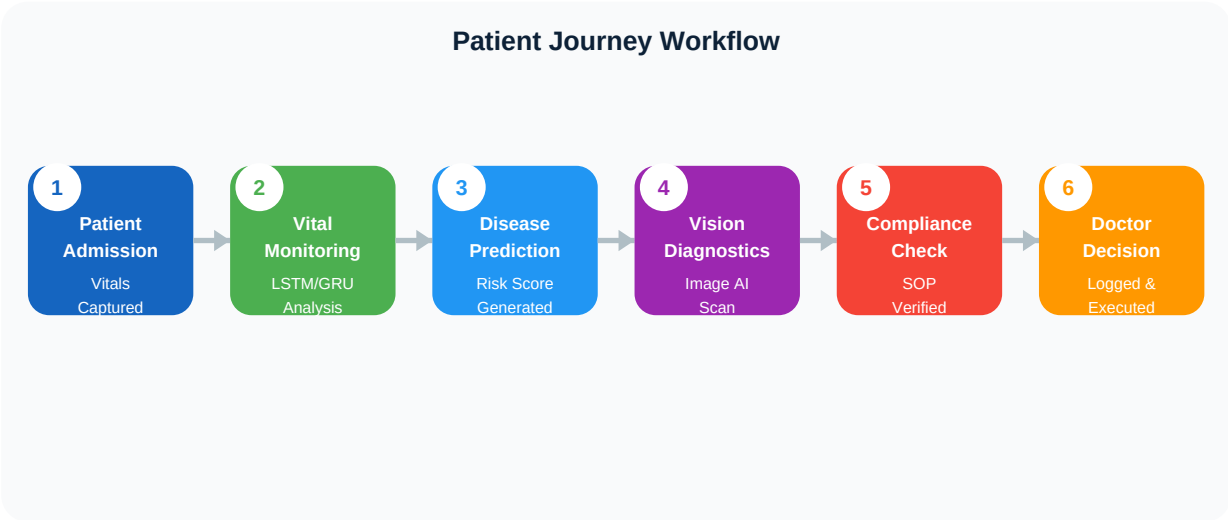


Figure 2 — End-to-End Patient Journey Workflow

Step 1 — Patient Admission

When a patient is registered, the system pulls their historical EHR data (prior diagnoses, allergies, medications) and opens a live monitoring session. Bedside IoT devices begin streaming vitals within seconds. A baseline risk profile is calculated immediately.

Step 2 — Continuous Vital Monitoring

The Vital Agent runs 24/7. Its LSTM/GRU model analyses rolling 30-minute windows of vital streams, comparing live data against each patient's personal baseline and clinical thresholds. Micro-alerts notify duty nurses before vitals become critical.

Step 3 — Disease Risk Prediction

Every 4 hours (or immediately after a vital anomaly), the Disease Prediction Agent re-evaluates risk. New lab results are automatically ingested and factored into the score. Any score above 75/100 automatically queues an ICU escalation request for compliance review.

Step 4 — Vision Diagnostics On Demand

When a physician orders imaging, the Vision Agent analyses the DICOM file within seconds of upload. The radiologist sees both the AI annotation overlay and the unmodified original image side-by-side to preserve clinical judgement.

Step 5 — Compliance Gate

Every action — ICU transfer, medication change, surgical booking — is held in a queue and evaluated by the Compliance Agent before execution. Missing paperwork, unsigned orders, or drug interaction flags block the action and display a clear reason on the dashboard.

Step 6 — Doctor Decision & Audit

The physician reviews the Supervisor Agent's consolidated recommendation, sees all supporting evidence, and approves or overrides with one click. Every decision — AI and human — is time-stamped and immutably logged in the EHR audit trail.

5. Compliance Agent — Decision Flow

The Compliance Agent is what transforms this into a **Safe AI** platform. No automated action bypasses it. The flow below shows how every AI recommendation moves through the verification gate before anything happens in the real world.

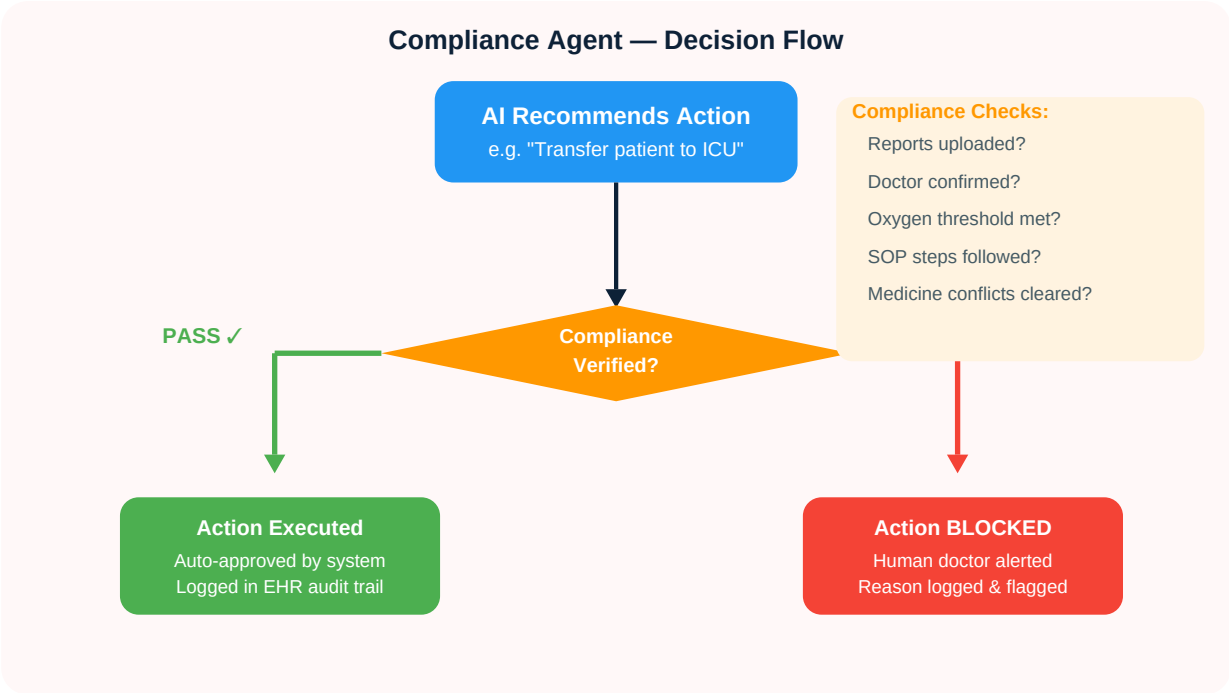


Figure 3 — Compliance Agent Decision Flow

Check Category	What It Verifies	If Check Fails
Documentation	All required reports uploaded to EHR	Action blocked, nurse notified
Doctor Authorisation	Physician has digitally signed the order	Alert to attending doctor
Threshold Validation	Vitals meet clinical criteria for action	Recommendation re-evaluated
Drug Interaction	No contra-indicated medication conflicts	Pharmacist alert triggered
SOP Sequence	All prerequisite steps completed in order	Missing steps listed on dashboard
Regulatory Policy	Complies with hospital & ministry guidelines	Compliance officer notified

6. User Experience — MediGuard AI Dashboard

Hospital staff interact with the system through **MediGuard AI**, a role-based web dashboard. Different personnel see tailored views: nurses see vitals and tasks, doctors see diagnostics and recommendations, compliance officers see audit logs. The interface uses real-time WebSocket updates so every change in a patient's condition appears instantly.

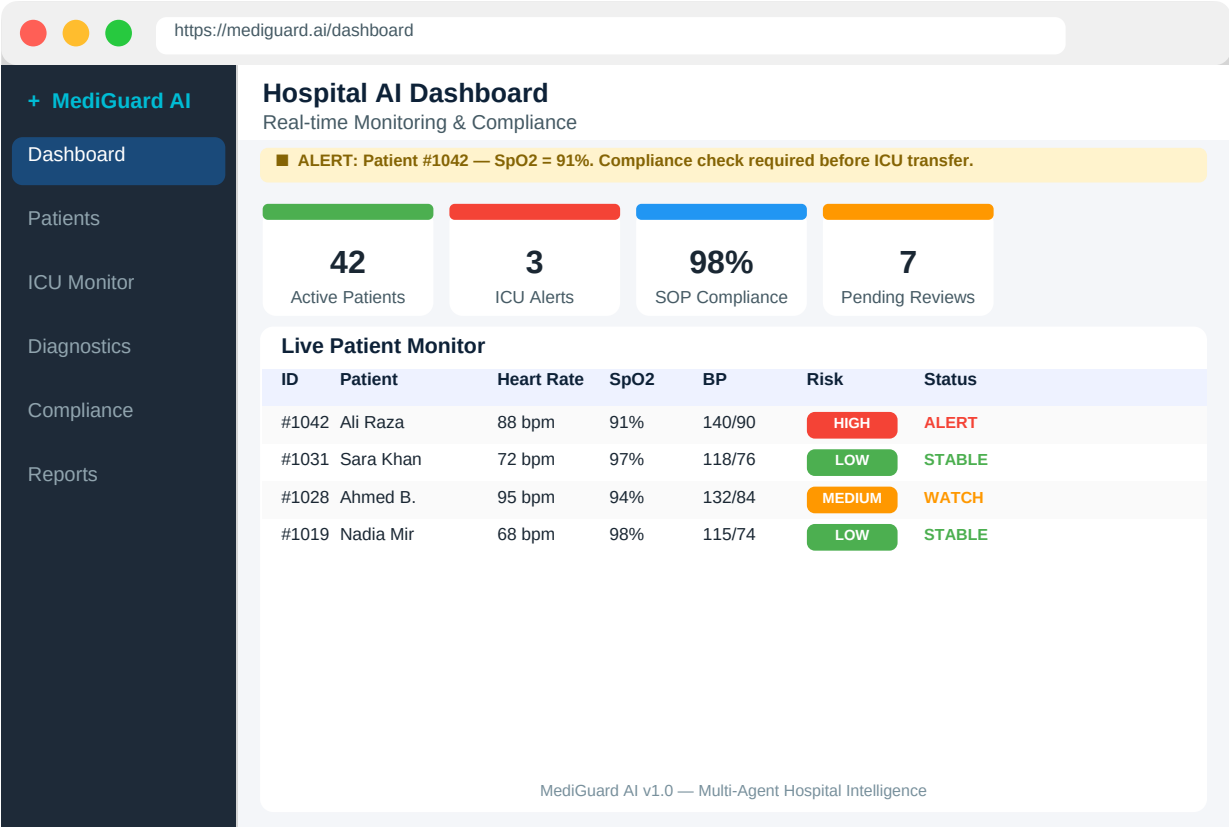


Figure 4 — MediGuard AI Dashboard (Physician View Mockup)

Role-Based Access & Views

User Role	What They See	Key Actions
Physician	Diagnostics, AI recommendations, risk scores	Approve/override, sign orders
Nurse	Live vitals, alert queue, task list	Log observations, upload reports
Radiologist	AI-annotated + original images side-by-side	Confirm/reject AI findings
Compliance Officer	SOP audit logs, violation reports, blocked actions	Investigate flagged events
Administrator	Bed occupancy, resource utilisation maps	Schedule staff, manage beds
IT / AI Team	Model performance metrics, concept drift alerts	Retrain, adjust thresholds

7. How All Functionalities Work Together

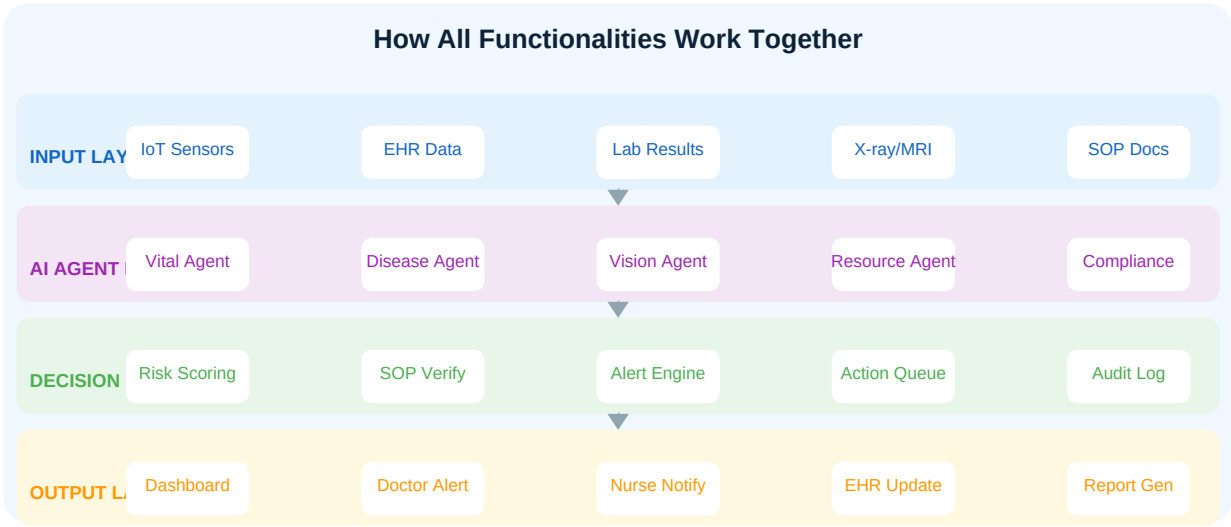


Figure 5 — Four-Layer End-to-End Data & Decision Flow

The four layers form a continuous loop. Sensor and EHR data feeds the AI agents, whose outputs drive decisions validated by compliance rules, and then delivered as dashboard updates, alerts, and audit records. The loop restarts with every new data event, making the system reactive in near-real time.

Live Integration Scenario

The table below traces a real patient event from admission to ICU transfer, showing which agent fires at each step:

Time	Event	Agent	Outcome
09:00	Patient Ali Raza (#1042) admitted	System	Monitoring session started
09:05	SpO2 drops to 91%	Vital Agent	Alert raised — Watch mode
09:10	Risk score jumps to 82/100	Disease Agent	ICU transfer queued
09:12	Compliance check initiated	Compliance Agent	Missing: Doctor sign-off
09:12	Doctor alerted via push notification	Supervisor Agent	Action held pending approval
09:16	Dr. Farhan approves ICU transfer	Human Decision	Transfer executed & logged
09:18	Chest X-ray ordered	Vision Agent	Preliminary: possible fluid
09:20	Radiologist confirms, EHR updated	Human + AI	Full audit trail recorded

8. Technical Stack

Layer	Technology	Purpose
Time-Series AI	LSTM + GRU	Vital sign anomaly detection & ICU prediction
Signal Analysis	1D-CNN	ECG heartbeat pattern classification
Vision AI	ViT-Base / MobileViT	X-ray, MRI, CT, skin disease classification
NLP / Policy	BERT fine-tuned	SOP document parsing & compliance checking
Tabular ML	XGBoost + Ensemble	Disease risk scoring from EHR data
Orchestration	LangGraph / AutoGen	Multi-agent workflow & message routing
Backend API	FastAPI (Python)	RESTful API for agent communication
Frontend	React.js + Tailwind CSS	Real-time dashboard with WebSocket updates
Database	PostgreSQL + TimescaleDB	Patient records & time-series vital storage
Messaging	Apache Kafka	Real-time event streaming between agents
Deployment	Docker + Kubernetes	Containerised microservices for each agent

9. Research Contribution & Academic Value

→ Novel Compliance Layer for Multi-Agent AI

Integrating a formal SOP verification engine into a multi-agent AI pipeline is a new research contribution. Most existing systems give recommendations — this system enforces governance before any action executes.

→ Cross-Domain Extension of Multi-Agent Architecture

The project directly extends an existing multi-agent agricultural AI framework into healthcare, proving the architecture's generalisability — a strong academic argument for the supervisor's existing work.

→ Human-in-the-Loop Safe AI Design

The system does not replace doctors; it augments them. This human-in-the-loop design aligns with WHO guidelines on AI in Health and current AI ethics research.

→ Reproducible & Benchmarkable

Using publicly available datasets (PhysioNet, CheXpert, MIMIC-III) makes results reproducible and directly comparable to published state-of-the-art literature.

Target Publication Venues	• IEEE Journal of Biomedical and Health Informatics • Applied Intelligence (Springer) • Journal of Medical Systems • AAAI / NeurIPS Healthcare Workshop
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10. What to Say to Your Supervisor

"Sir, instead of building a generic AI assistant, we want to extend your multi-agent framework into the healthcare domain while adding a formal governance and compliance layer — demonstrating that AI can be both intelligent and safe. We specifically need your expertise in GRU/LSTM time-series analysis, Vision Transformer models, and multi-agent coordination, as these form the backbone of our proposed architecture."

11. Conclusion

The **Smart Hospital Agentic AI** project is a technically rigorous, well-scoped, and academically publishable Final Year Project. It addresses a real gap in healthcare AI — powerful but ungoverned AI systems — by integrating compliance and governance directly into the multi-agent pipeline. The architecture aligns closely with the supervisor's existing expertise in LSTM/GRU modelling, Vision Transformers, and multi-agent orchestration, making collaboration natural. The combination of time-series AI, vision transformers, and rule-based compliance verification across five specialised agents represents a genuinely novel contribution to the field of safe AI in healthcare.